Certiva — Methodology & Benchmark Results Summary

Hybrid Search v3 · Tested September 11, 2026 · This document is freely accessible — link directly, no lead-gen form required.

1. Methodology

Test environment: Windows-11-10.0.26200-SP0, Python 3.14.7. Corpus: 13 core regulations, manually verified against official sources (JDIH BPK/Ministry of Finance) by a non-commercial network of legal practitioners (10 notary offices + 1 law firm). Article and cross-reference relation counts are not listed as static figures here — these are computed live from the database & graph via the **corpus_stats** MCP tool (see *Certiva — MCP Tooling Summary*, Section 4), so they always stay consistent with the corpus's current state at every call.

Definition of “first-try correct” (Precision@1): the top-ranked regulation returned by the engine exactly matches the regulation manually established as the correct answer for that query. Recall@5: the correct regulation appears among the top 5 results.

Test set: 75 real-world regulatory scenarios (7 categories — see Section 3) + 28 guardrail queries (outside the legal domain, must return empty results). Scenarios were built & verified by the same legal practitioners as the ground-truth network above.

Reproducibility: this report is generated automatically by a benchmark script that auto-stamps the run date & code version on every execution, ensuring published figures are always tied to the exact system version that produced them.

2. Headline Results

97.3%	98.7%	15.1 ms	96.4%
Precision@1	Recall@5	Latency	Guardrail Pass
(73/75)	(74/75)	Average	(27/28)

All figures are taken directly from the production benchmark — not estimates. RAM after the embedding model is loaded: 798.2 MB.

3. Breakdown by Test Category — Hybrid vs BM25-only

Category	N	Hybrid R@5	Hybrid P@1	BM25-only R@5	BM25-only P@1	Interpretation
semantic_recovery	1	0/1	0/1	0/1	0/1	The only pure dense-only test; not yet validated
disambiguation	6	6/6	6/6	6/6	6/6	Hard test: choosing among closely related regulations
regression	31	31/31	31/31	31/31	31/31	No-regression check (BM25 was already correct before)
regression_akronim	7	7/7	7/7	7/7	7/7	Single-word acronyms must not be blocked by the gate
must_pass_akronim	14	14/14	13/14	14/14	13/14	False-negative check: legitimate acronyms must return non-empty results
must_pass_frasa	10	10/10	10/10	10/10	10/10	False-negative check: specific short phrases must return non-empty results (up from 9/10 — see Section 6)

Category	N	Hybrid R@5	Hybrid P@1	BM25-only R@5	BM25-only P@1	Interpretation
must_pass_regulasi	6	6/6	6/6	6/6	6/6	Direct regulation numbers must return non-empty results

> **Methodology note:** most categories above are regression checks (ensuring hybrid doesn't break cases that already worked in pure BM25), not independent proof of the semantic layer's benefit. The only pure dense-only test (semantic_recovery) has not yet succeeded — see Section 4.
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> **Note on corpus figures:** all corpus-size figures (article count, article-level relations, document-level relations) cited in any external material — website, deck, or this document — come from a single source: the **corpus_stats** tool on the Legal-Hybrid-Search server, computed live from the database & graph each time the tool is called, not static figures copied between documents. This is consistent with the policy stated in *Certiva — MCP Tooling Summary*, Section 4.

4. Limitations, Disclosed Honestly

3 items still open:

- **1 Precision@1 miss from a pure dense-only test.** A purely semantic query (“if a startup app has a bug, who can be sued?”) — empty result, both R@5 and P@1 failed. Status: a knowingly accepted limitation — we chose not to loosen the out-of-domain query filter just for this one case.
- **1 acronym ambiguity** (not a failure; two equally valid answers). “PPN” (VAT) — correct regulation lands in the top 5, not #1.
- **1 residual guardrail leak** out of 28 out-of-domain queries (pass rate 27/28 = 96.4%). “OTG for phone data transfer” → incorrectly surfaced UU-27-2022 (Indonesia's Personal Data Protection Law). Root cause: a collision between a legitimate technical term (“data transfer” is a term used in the PDP Law) and everyday meaning (USB OTG) — still under evaluation whether such queries are actually relevant to surface (not a pure bug).

Items closed since the previous version:

- 4 guardrail leaks reported in the previous version are now closed (how to grow chili peppers in a pot, how to play chess, how to fix wifi, online police clearance certificate) via improved generic-word filtering in the keyword search layer — validated with 0 regressions across all 103 test scenarios.
- 2 Precision@1 misses reported in the previous version are now down to 1. The “PPh 22 Pihak Lain” (withholding tax) case was closed via an improvement to the keyword & semantic result-merging algorithm, without introducing new regressions (re-validated against all 103 scenarios, including 3 other scenarios that were temporarily affected during an earlier fix iteration and have since been restored).

Status of the semantic layer: its independent contribution (without BM25) is still under active validation — the only pure dense-only test in this test set has not yet succeeded. We report this transparently, rather than hiding it.

5. Reproducibility & Contact

Full scenario & guardrail query logs, along with test scripts, are available on request for judges and investors. This is an internal benchmark — a self-reported team result; independent verification is always welcome. Contact **certiva@arsiva.tech** or visit **arsiva.tech**.

6. Revision History

The materials we registered for CoCreate Pitch 2026 used benchmark data as of **September 3, 2026**: Precision@1 72/75 (96.0%), Guardrail Pass 23/28 (82.1%).

Since then, the team has continued an ongoing cycle of fixes & re-testing. This document reflects the **LATEST results as of September 11, 2026**, with figures:
• Precision@1: 72/75 → **73/75**
• Guardrail Pass: 23/28 → **27/28**

We chose to publish this updated version transparently — including recording changes since the registration version — because we believe a tested & documented improvement process is part of the value we want to show, not something to hide.